

Vatsal Jha

vatsaljha17@gmail.com · LinkedIn · GitHub · Google Scholar · vatsaljha.dev

Bangalore, India

Selected Software & Systems Experience

CtrlAltHeal

Feb–May 2026

Freelance Platform Engineer, Health-tech

Remote

- **Sole engineer** of two parallel backend stacks (Node/Express/Prisma diet platform; separate C# ASP.NET/EF Core management platform) behind 4 client surfaces; built DOCX/XLSX/CSV pipelines that reconcile legacy nutrition/PCOS clinical documents into structured data via Jaccard fuzzy matching, plus architecture/security/runbook handover docs.

testAIng

Jan–Jul 2026

Machine Learning Intern

Remote

- Hardened model serving with Service/HPA/KEDA, readiness/liveness probes, rolling deploy/rollback and metrics; implemented meeting automation with speaker/transcription models.
- Also owned R&D detecting AI-generated/deepfake speech (ECAPA-TDNN, WavLM, mixture-of-experts) (**0.94–0.96 AUC**; **1.97% EER** after Hindi fine-tuning); **1st Prize/70+ interns (research presentation competition)**. [blog]

Stack: FastAPI, Docker, Kubernetes, Prometheus/Grafana, Playwright/Xvfb, WhisperX, Pyannote.

Samsung Research India

May–Jul 2025

Software Engineering Intern, Server Platform / MAGPIE

Bangalore

- Individually **designed and built distributed-service RCA from scratch**: ADK orchestration + MCP interfaces; cross-incident persistence/retrieval, structured validation and guarded diagnosis/reporting.
- Built a fault-injection/evaluation environment against a published **90-scenario** microservice fault-injection benchmark (9 fault types × 10 incidents); owned implementation, integration, evaluation and handoff; strong Top-1 root-cause identification on an internal dev slice; received a full-time **Pre-Placement Offer (PPO)**.

Stack: Google ADK/MCP, Kubernetes, Prometheus, Jaeger, OpenSearch, Milvus, OpenTelemetry, Chaos Mesh.

CEEW

Jun–Aug 2024

Data Science Intern

Delhi

- Built a modular Python/Streamlit solar+BESS tool covering dispatch, tariffs, outages, battery-sizing search and multi-year financial metrics; also shipped a public NLP decision-support prototype over 93K skill/task segments that later informed the public **CEEW–ADB SMART** platform. [solar] [SMART]

Education

Vellore Institute of Technology

2022–2026

B.Tech. Computer Science & Engineering (IoT)

Vellore, India

CGPA **9.23/10**; **Top 1%** nationally, NPTEL Reinforcement Learning (IITM) – invited to interview for RL Instructor, IITM BS; **Top 5%** Responsible & Safe AI (IIITH × IITM). **Final Year Project**: SOPRA (see below).

Systems & Product Engineering

Eldermere | Go, SvelteKit, PostgreSQL, WebSockets, Docker

[code] [live] [docs]

- Built and publicly deployed multiplayer browser MUD with room/presence WebSocket state, persistent sessions/player state, JSON content packs, validation/tests, Docker Compose and procedural visual/ambient-audio systems.

ChatterBot | FastAPI, Ray Serve/vLLM, Redis, RabbitMQ, PostgreSQL, SSE

[code]

- Implemented sync, queued async and streaming inference modes with Nginx/Docker, context caching, AWQ path, rate limiting, retries/circuit breakers and Locust load tests; architecture designed for horizontal scaling.

Sahayata | FastAPI, ADK/MCP, Gemini Live, Vertex AI RAG, WebSockets, Mem0

Google GenAI Exchange [org]

- Built FastAPI backend and MCP tools for a voice-first wellness prototype with Gemini Live, ADK agent routers, and Mem0 persistent memory (designed for wearable integration); presented working demo.

KSP Traffic Simulation | Python, OSM, SUMO/TraCI, simulation automation

[code]

- Built automatic location→OSM→SUMO/TraCI simulation/evaluation pipeline for traffic-flow optimization; **Top-3/380 teams** in Karnataka State Police Datathon.

AI / Research Engineering

HOMA | Multi-agent orchestration, edge SLMs, multilingual systems

[code]

- Co-led 4-person system decomposing multi-device commands into sequential/concurrent execution across **11+ languages**; presented at ICCNT 2025; **Samsung PRISM Excellent Worklet Award**.

CA-AMFA + EV SoC | IoT/security + telemetry ML

[CA-AMFA] [EV paper]

- CA-AMFA: Raspberry Pi adaptive-authentication prototype with Dr. Yokesh Babu S, first-author **IEEE Best Paper**. EV SoC: test **MSE ≈ 0.041** from a stochastic trip-segmentation ML pipeline, advised by Dr. B. Ashok under a Royal Academy of Engineering grant; *Journal of Energy Storage* (IF ≈ 11, Q1) 97 (2024). [CA code] [EV code]

Manuscripts under review:

- **SOPRA**, LLM Java test repair via oracle localization and SUT-diff evidence routing; **58.7% exact match (+14.7 pp)**. *SANER 2027*.
- **BehaviorFL/BehaviorFL-MT**, privacy-preserving federated in-vehicle recommendation study under heterogeneous sensors, co-inventor (Published Patent App. **202641089937**); discovery-aware sampling, **5.8×** rare-task exposure at fixed budget. *NeurIPS 2026 NewInML Workshop*.

Additional Engineering & Recognition

Hackathons: 2x National Winner (IISF 2023 Space Hackathon for Narad 22-language assistant [code]; Samsung PRISM for recommender system [code]).

Open-Source: PromptTrend API [code] [PyPI]. **Leadership**: *Co-Founder, Team Prometheus* (30+) [site]; led zero-to-one RoboCup AI R&D; HINT'25 NMPC paper [code]. **Vice Chair→Chair, Youth Red Cross VIT** (750+ donations).

Languages: Python, Go, JS/TS, Java, C++, C#, SQL. **Systems**: React/React Native, Node, ASP.NET, Docker/K8s, Ray/vLLM, Redis/PostgreSQL, WebSockets, MCP/ADK. **ML**: PyTorch/HF, LLMs/agents, RAG, RL.